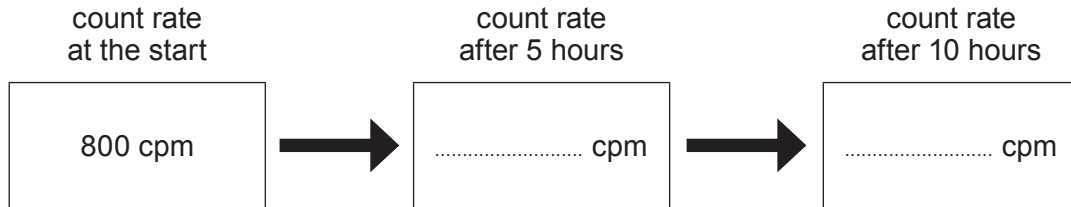
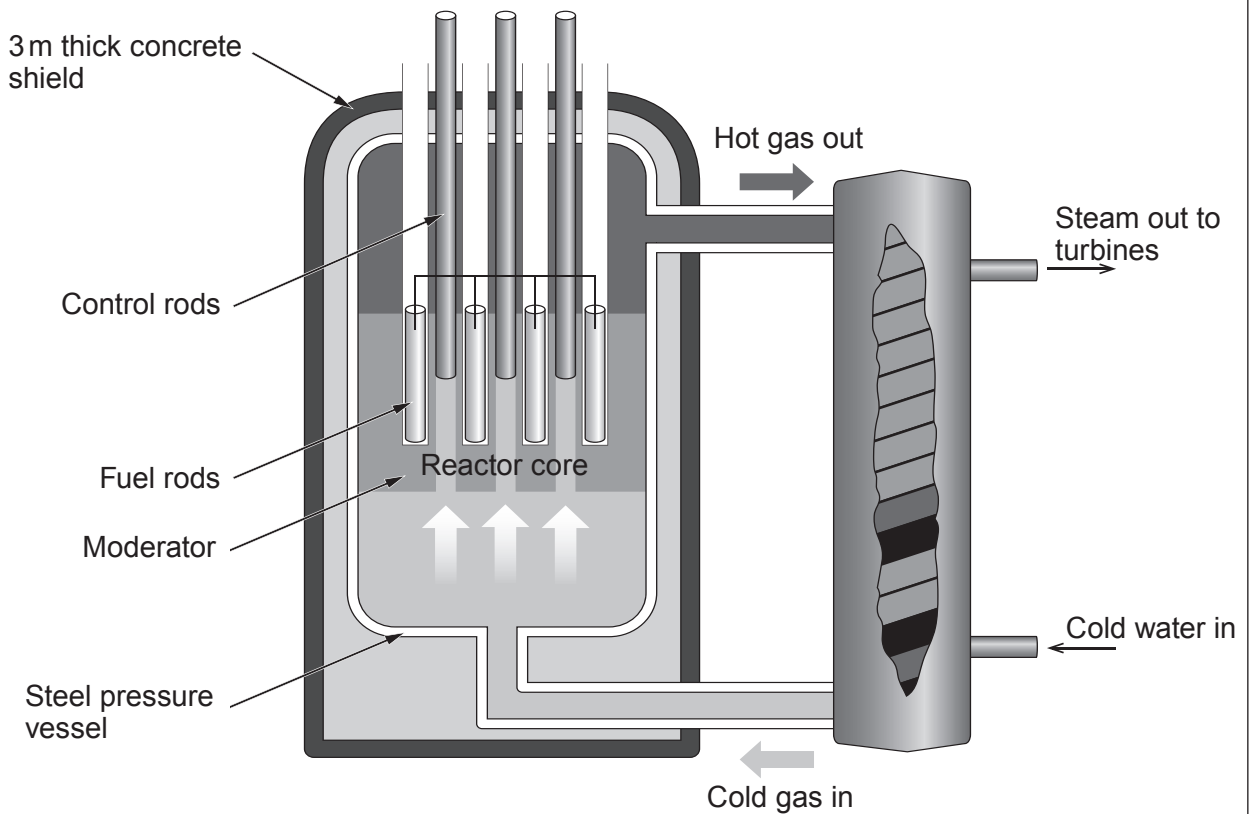


2. The half-life of a radioactive substance is the time taken for its activity to reduce by half.

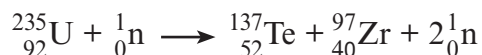
- (a) **Complete the following diagram** to show the count rate at the times shown.
The substance has a starting count rate of 800 counts per minute (cpm) and a half-life of 5 hours. [2]



- (b) The following diagram shows a nuclear fission reactor in which the uranium fuel is contained inside a steel pressure vessel which is surrounded by a 3m thick concrete shield.



A uranium nucleus can split apart, forming two smaller nuclei in a nuclear reactor. One possible nuclear reaction is shown below in which a nucleus of uranium forms nuclei of tellurium (Te) and zirconium (Zr).



- (i) State the number of neutrons released in this reaction. [1]
- (ii) Complete the following sentences about the nuclear reactor by underlining the correct word or phrase in each bracket. [4]

A nucleus of uranium is made to split into two smaller nuclei when it captures **(one / two / three)** neutron(s).

For fission to occur, the neutron(s) to be captured must be **(fast moving / stationary / slow moving)** and this takes place in the **(moderator / control rods / fuel rods)**.

An uncontrolled chain reaction is prevented by using **(control rods / a moderator / concrete shielding)**.

- (c) State **two** reasons why it is difficult to store waste material from nuclear power stations. [2]

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